

|                        |                                                    |
|------------------------|----------------------------------------------------|
| <b>Iniziato</b>        | Thursday, 13 January 2022, 15:12                   |
| <b>Stato</b>           | Completato                                         |
| <b>Terminato</b>       | Thursday, 13 January 2022, 15:41                   |
| <b>Tempo impiegato</b> | 29 min. 29 secondi                                 |
| <b>Punteggio</b>       | 13,00/15,00                                        |
| <b>Valutazione</b>     | <b>26,00</b> su un massimo di 30,00 ( <b>87%</b> ) |

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following is not a property of a *metric* distance function

Scegli un'alternativa:

- ☒ a. Boundedness
- ☐ b. Symmetry
- ☐ c. Triangle inequality
- ☐ d. Positive definiteness

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Given the two binary vectors below, which is their similarity according to the Simple Matching Coefficient?

**abcdefghij**

1000101101

1011101010

Scegli un'alternativa:

a. 0.2

b. 0.3

c. 0.1

✓ ~~d. 0.5~~

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following statements is *true*?

Scegli una o più alternative:

a. The noise always generate outliers

b. The data which are similar to the majority are never noise

✓ ~~c. The noise can generate outliers~~

✓ ~~d. Outliers can be due to noise~~

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

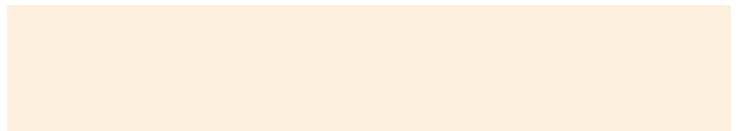
Given the definitions below:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

which of the formulas below computes the accuracy of a binary classifier?

**Scegli un'alternativa:**

-  a.  $TN / (TN + FP)$
-  ~~b.  $(TP + TN) / (TP + FP + TN + FN)$~~
- c.  $TP / (TP + FP)$
- d.  $TP / (TP + FN)$



Domanda 5

Risposta errata

Punteggio ottenuto 0,00 su 1,00

What is the *cross validation*

**Scegli un'alternativa:**

- ☒ a. A technique to obtain a good estimation of the performance of a classifier with the training set
- ☐ b. A technique to improve the quality of a classifier
- ☒ c. A technique to obtain a good estimation of the performance of a classifier when it will be used with data different from the training set
- ☐ d. A technique to improve the speed of a classifier

Domanda 6

Risposta errata

Punteggio ottenuto 0,00 su 1,00

A Decision Tree is...

**Scegli un'alternativa:**

- ☐ a. A tree-structured plan of tests on single attributes to forecast the cluster
- ☒ b. A tree-structured plan of tests on multiple attributes to forecast the target
- ☒ c. A tree-structured plan of tests on single attributes to forecast the target
- ☐ d. A tree-structured plan of tests on single attributes to obtain the maximum purity of a node

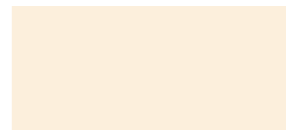
Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

When training a neural network, what is the *learning rate*?

- a. The speed of convergence to a stable solution during the learning process
- b. The ratio between the size of the hidden layer and the input layer of the network
- c. The slope of the activation function in a specific node
- ☒ d. A multiplying factor of the correction to be applied to the connection weights



Domanda **8**

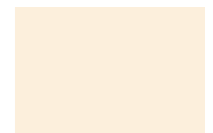
Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What measure is maximised by the Expectation Maximisation algorithm for clustering?

**Scegli un'alternativa:**

- a. The likelihood of an example
- b. The likelihood of an attribute, given the class label
- c. The support of a class
- ☒ d. The *likelihood* the distributions, defined by the parameters found, given the data available



Domanda 9

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What does K-means try to minimise?

Scegli un'alternativa:

- ☐ a. The *separation*, that is the sum of the squared distances of each point with respect to its centroid
- ☐ b. The *separation*, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset
- ☒ c. The *distortion*, that is the sum of the squared distances of each point with respect to its centroid
- ☐ d. The *distortion*, that is the sum of the squared distances of each point with respect to the points of the other clusters

Domanda 10

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following characteristic of data can reduce the effectiveness of DBSCAN?

Scegli un'alternativa:

- ☒ a. Presence of clusters with different densities
- ☐ b. All the variables are the same range of values
- ☐ c. Presence of outliers
- ☐ d. Clusters have concavities

Match the rule evaluation formulas with their names

✓ 
$$\frac{1 - \text{sup}(C)}{1 - \text{conf}(A \Rightarrow C)}$$
 conviction

✓ 
$$\text{sup}(A \cup C) - \text{sup}(A)\text{sup}(C)$$
 LEVERAGE

✓ 
$$\frac{\text{conf}(A \Rightarrow C)}{\text{sup}(C)}$$
 LIFT

✓ 
$$\frac{\text{sup}(A \Rightarrow C)}{\text{sup}(A)}$$
 CONFIDENCE

$$\frac{conf(A \Rightarrow C)}{sup(C)}$$

→ Lift,

$$\frac{sup(A \Rightarrow C)}{sup(A)}$$

→ Confidence

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Consider the transactional dataset below

### ID Items

- 1 A,B,C ✗ -
- 2 A,B,D
- 3 B,D,E
- 4 C,D
- 5 A,C,D,E ✓

Which is the *confidence* of the rule  $A,C \Rightarrow B$ ?

Scegli un'alternativa:

- ☐ a. 100%
- ☐ b. 20%
- ☐ c. 40%

✓ ☒ d. 50%

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a dataset with  $D$  attributes, how many subsets of attributes should be considered for feature selection according to an exhaustive search?

Scegli un'alternativa:

- ☐ a.  $O(D!)$
- ☐ b.  $O(D)$
- ☐ c.  $O(D^2)$
- ☒ ~~d.  $O(2^D)$~~

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the **coefficient of determination  $R^2$** ?

- ☐ a. Measure the amount of error in a linear regression model
- ☒ b. An index of goodness for a classification model
- ☒ ~~c. Provide an index of goodness for a linear regression model~~
- ☐ d. Measure the amount of error in a regression model

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

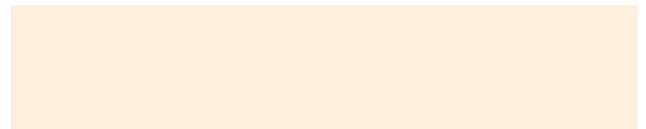
Which is different from the others?

**Scegli un'alternativa:**



Decision Tree

- b. Dbscan
- c. Expectation Maximisation
- d. K-means





Vai a... 